FIRECHRONICLE

FINAL TEAM RULEBOOK — PARÁ, BRAZIL

NASA Space Apps Challenge 2026 | Harmonization of MODIS and VIIRS Hot Spots

Version: Pará release amendment v1.0 | Effective: 10 November 2026 | Integration target: 12 November 2026

Team: 6 members | Final submission: 13 November 2026, 6:30 PM (Dhaka time, confirm with portal)

One mission

Build a real, map-first historical burning-activity explorer for Pará state in Brazil. Translate supported pre-VIIRS Aqua MODIS observations into estimated VIIRS S-NPP-equivalent hotspot detection counts, use observed S-NPP VIIRS counts for the modern era, and let users select areas within Pará to explore historical patterns, seasonality and model evidence.

Binding change from the previous rulebook

Pará (PA), Brazil, replaces the previous United States / Brazil / Argentina geographic scope. The website is Amazon-themed but does not claim to cover the entire Amazon biome or all of Brazil. Every raw detection is spatially filtered against the agreed Pará state polygon before being included in Pará analyses. All other scientific integrity rules and team handoffs remain in force.

This is the implementation source of truth. Change shared scientific definitions or interface contracts only to fix a verified error or blocker, with affected owners and the collaboration lead agreeing.

1. Locked project decisions

Decision	Binding team standard
Release geography	Pará state (PA), Brazil only. The map can show geographic context outside Pará, but clickable analytics are enabled only inside the agreed Pará study boundary.
Selected area	Users can click supported analysis blocks, draw/select an area inside Pará, and inspect municipalities or presets only where their boundaries are verified.
Measurement	VIIRS S-NPP-equivalent hotspot detection counts. NOT individual fire events, burned area, fire intensity, deforestation, or fire-spread forecasts.
Reference sensors	Aqua MODIS as historical predictor and overlap input; VIIRS S-NPP as calibration target and modern observed reference.
History	Mid-2002–2011: estimated using historical Aqua MODIS where supported. 2012 onward: observed S-NPP where available; latest NRT records only if labeled Provisional.
Spatial system	EPSG:6933 equal-area. Base cells 10,000 m × 10,000 m, origin (0,0), floor-indexed. Test 50 km blocks for monthly model calibration.
Models	Ratio-of-totals mandatory baseline; Negative Binomial regression as main candidate. Other models only after working integration.
Website	Map-first Pará explorer, year × month burning calendar, seasonal charts, Sensor Lab, optional scripted CHRONO guide.
Backend	Preserve the chosen working database/query architecture; FastAPI /api/v1 + Next.js/TypeScript + MapLibre. Parquet is the interchange format.

Release promises

• A user can choose a location inside Pará and receive actual available historical analysis, not a decorative static map.

• Every displayed value identifies Observed / Estimated / Provisional / Not processed / Unsupported and the relevant time period.

• No map cell, month, or location is filled with invented VIIRS detections or silently assigned zero to disguise missing observations.

2. Data acquisition and Pará geographic filtering

Study area definition: the official administrative polygon of Pará state, Brazil, from IBGE state territorial boundaries (Malhas Territoriais). Store the boundary, source URL, release year, CRS and processing date in version control or the data manifest. Use one frozen boundary version for the release. Do not use a rectangular bounding box as the final Pará membership test.

Priority	Data Engineer acquisition	Purpose
P0	Brazil FIRMS country-year Aqua MODIS and VIIRS S-NPP CSVs for multiple overlap years, emphasizing 2012–2020; immediately clip each to Pará.	Paired monthly calibration, year and geographic holdouts.
P0	Brazil Aqua MODIS for at least one historical year before 2012, then expand backward to mid-2002 where available; clip to Pará.	Demonstrate a meaningful pre-VIIRS historical calendar.
P1	All available annual Brazil MODIS and S-NPP files needed to fill the Pará historical calendar.	Longer calendar and stronger temporal validation.
P2	S-NPP NRT for recent dates if standard-processing files do not yet cover them.	Optional recent activity, marked Provisional.

Canonical point-in-polygon procedure

• Read source longitude and latitude; create geographic points in EPSG:4326.

• Load the frozen Pará IBGE polygon; transform point and polygon layers to a common CRS.

• Retain points within/intersecting the Pará boundary, using one documented boundary-edge policy. Remove spatial-join helper columns from output.

• Apply the identical Pará membership test to Aqua MODIS, S-NPP VIIRS, all training periods and historical inference periods.

• Audit the filtered subset: compare raw Brazil row counts against retained Pará row counts, test known interior/exterior coordinates, check latitude/longitude order, and preserve rejected records separately.

Crucial: Pará is not the entire Amazon. A record in Amazonas, Mato Grosso or Maranhão must not enter Pará statistics merely because it is near the rainforest. Also, the national FIRMS archive is a source container, not the project analysis region.

Canonical grid assignment

After geographic filtering, transform coordinates to EPSG:6933 meters; compute ix = floor(x / 10000), iy = floor(y / 10000), grid_id = "ix_iy". Derive block_id from floor(ix / 5) and floor(iy / 5). This creates nested 50 km calibration blocks from 10 km cells. The Data Engineer owns this tested function; do not replace it with a 0.1-degree latitude/longitude approximation.

At state boundaries, aggregate only actual hotspot points inside Pará. If a 50 km block crosses Pará’s border, report the model domain and clipped support correctly; do not claim that the full block lies inside Pará.

3. Measurement, time and coverage rules

Era	Value shown	Non-negotiable label
Mid-2002–2011	Estimated S-NPP-equivalent counts from historical Aqua MODIS inputs when validated support permits.	Estimated · historical extrapolation
2012–latest finalized archive	Actual observed VIIRS S-NPP detection counts; overlap MODIS is used for model training/validation but not added to display totals.	Observed · S-NPP
NRT-only recent dates	S-NPP near-real-time detections only if pipeline supports them and provenance is retained.	Provisional · NRT

• Never add MODIS counts and S-NPP counts to form the harmonized total.

• Show a visible transition at the beginning of the S-NPP observation record in 2012; the historical and modern series are on an intended reference scale but are not magically identical.

• A missing fire-positive detection row is not proof of a clear-sky observation or a verified zero. Only label 0 as verified where usable observation evidence supports it.

• Monthly aggregation simplifies matching, but does not correct clouds, overpass exposure or coverage. If FIRMS positive-detection CSVs are the only evidence, mark the model as coverage-unadjusted/preliminary.

• Keep source dataset, spacecraft, processing version, UTC acquisition time, fire-type availability, confidence, FRP, latitude and longitude in the provenance data.

• Verify fire-type meaning by product/version before filtering. Do not invent a brick-kiln classifier or a NOAA-20 gap-fill conversion for the release without validation.

• Hotspot count means recorded fire-positive satellite pixels/detections, not a count of distinct wildfires.

Comparing activity across years

Show a year × month calendar with explicit missing values. Display estimated historical values differently from observed values (for example hatching). Prefer season-matched comparisons using like-for-like measurement tiers. Mixed-era anomalies or comparisons must be gated on demonstrated calibration stability. Never generate arbitrary confidence bands.

4. Calibration, model selection and validation

Modeling unit: Pará-filtered 50 km block × year × month × day/night. Training input M = Aqua MODIS recorded detection count; target V = S-NPP VIIRS recorded detection count. This is a pragmatic monthly observed-detection calibration, not an assertion that zero detections equal verified zero observed fires.

Candidate	Implementation / decision
Model 0 — mandatory	Ratio of totals: k = sum(V) / sum(M), learned within each training fold and applied to M. It cannot predict missed S-NPP detections where M = 0.
Model 1 — primary candidate	Negative Binomial count regression; candidate predictors log(1+M), month as season/categorical, day/night, and geographic/block terms only if identifiable and robust.
Optional	Linear Regression or XGBoost only after the baseline, NB validation and integration are working.

Validation that determines whether the model can be shipped

• Train on multiple MODIS–VIIRS overlap years within Pará. Prefer 2012–2020 for initial orbit-consistent calibration; treat later Aqua drift periods with caution.

• Hold out complete years and independent geographic blocks within Pará. A random row split is not sufficient.

• Evaluate Poisson deviance, mean absolute error, aggregate count bias, seasonal pattern/peak-month agreement and especially geographic-block bias. Report the number of eligible units and any excluded records.

• Compare all reported models on the same test data; deploy Negative Binomial only if it improves materially over the simple baseline.

• Test M = 0 cases where independently supported; an NB nonzero estimate for zero MODIS detections is not itself evidence that missing fires were recovered.

• For a held-out overlap year, predict its VIIRS-equivalent counts using only MODIS predictors, then chart predicted vs actual S-NPP. This is the key Sensor Lab evidence.

• Do not automatically apply a Pará-trained model to the entire Amazon or Brazil. Any broader-area estimate requires geographic validation first.

Spatial precision rule: A model fitted and validated at 50 km should output 50 km estimated blocks. The observed point layer can retain original coordinates and the observed aggregate layer can use 10 km cells. Never arbitrarily downscale 50 km estimates and claim validated 10 km results.

5. Data Scientist → Backend B exact handoff

Member 2 (Data Scientist) owns training, validation, historical inference and a callable Python interface. Member 4 (Backend B) owns the FastAPI endpoint and HTTP error handling. The model author does not need to deploy a second API service.

Deliverable	Required behavior
model.joblib / package	Selected trained model and identical inference preprocessing; reproducible dependencies.
predict.py	predict_viirs_equivalent(...) and batch inference; reject unsupported geography, dates and feature values.
model_info.json	Version, target=VIIRS S-NPP-equivalent hotspot detections, region=Pará PA Brazil, supported years, features, 50 km calibration scale and limitations.
validation_metrics.json	Held-out year and geographic-block metrics, baseline comparison, eligible sample counts and assumptions.
estimated_viirs_block_month.parquet	Historical estimates keyed by region_id, block_id, year, month, daynight; retain estimation and coverage statuses.
examples + tests	example_input.json, example_output.json, requirements.txt, model-card.md and one reproducible smoke test.

One shared model response schema

Required: region_id="BR-PA"; block_id; year; month; daynight; estimated_viirs_count (number or null); value_type="estimated"; prediction_status; coverage_status; model_version; measurement_unit="VIIRS_S-NPP-equivalent_detection_count"; calibration_scale_km=50. Validated intervals are optional, not invented.

Status values: ok_preliminary, unsupported_region, unsupported_period, out_of_training_range, insufficient_coverage, missing_required_feature, model_error. Backend must preserve null and exact status; never substitute zero.

Illustrative response — numbers are fictional

{ "region_id": "BR-PA", "block_id": "123_-54", "year": 2005, "month": 9, "daynight": "D", "estimated_viirs_count": 61.4, "value_type": "estimated", "prediction_status": "ok_preliminary", "coverage_status": "unknown", "model_version": "example-v0", "calibration_scale_km": 50 }

Backend primarily reads precomputed monthly estimates for map, charts and calendar. Do not retrain or recompute the full archive on a map click.

6. Shared storage, API and frontend contracts

Storage: Keep the already functioning PostgreSQL/PostGIS or DuckDB/Parquet path; do not introduce a second incompatible backend. Parquet/CSV is the team exchange format. Backend B exposes stable FastAPI /api/v1 routes and shares JSON fixtures immediately.

Endpoint	Pará release requirement
GET /api/v1/regions	Return Pará BR-PA only as active region, its polygon/bounds, processed years and statuses; Brazil is geographic context, not a second supported analysis region.
GET /api/v1/harmonized	Return 50 km estimated block values or separately identified observed summaries, with year/month and type/status.
GET /api/v1/timeseries	Monthly values for supported user-selected area within Pará, keeping genuine data gaps.
GET /api/v1/calendar	Year × month burning calendar plus optional daily drilldown and data type labels.
GET /api/v1/hotspots	Actual reported hotspot point locations in Pará, with source metadata; paginate and bound queries.
GET /api/v1/model-info	Selected baseline/model, held-out results, calibration units, region=Pará, uncertainty and limitations.
GET /api/v1/coverage	Availability for Pará per year/month, unknown/verified status and unsupported reasons.

Frontend experience — Amazon storytelling, Pará analytical coverage

• Map: center on Pará within the Brazilian Amazon. Render a clear Pará boundary. Areas outside it can remain on the basemap but cannot generate fabricated analysis.

• Area selection: click 50 km modeled blocks; optionally draw a polygon inside Pará or choose verified municipalities. Explain partial-border and spatial-resolution limitations.

• Time Machine: year × month calendar, linked seasonal median/IQR (where comparable), selected-month detail and daily breakdown if available.

• Sensor Lab: compare held-out MODIS-based estimate with actual S-NPP counts; show real model metrics and honest limits.

• CHRONO: one optional scripted guide reacting to selected region, year, cell and quiz events; no invented scientific findings or generative model service.

• Interaction: clicking a map area updates the calendar and trends; selecting time updates map layers. Estimated/Observed/Provisional statuses must be consistent across views.

7. Six team members — responsibilities and handoffs

Owner	Owns	Must hand over
1 — Data Engineer	Brazil country-year downloads, IBGE Pará polygon, spatial clip QA, spacecraft/provenance checks, common EPSG:6933 grid, 50 km monthly tables, availability/coverage.	Filtered Pará datasets, overlap table, historical input table, observed S-NPP aggregates, clipping tests and data dictionary.
2 — Data Scientist	Ratio baseline, NB model, temporal and spatial validation within Pará, model guardrails, historical estimates and scientific report.	Saved model, predict.py, metadata, model card, held-out metrics and predicted Pará history.
3 — Backend A	One working query layer, geometry lookup, within-Pará area selection, aggregate queries and block/cell mapping.	Tested geo/time queries, data tables and map geometries.
4 — Backend B	FastAPI /api/v1 contracts, model inference integration, statuses, schemas, mock fixtures and endpoint tests.	Documented real APIs and consistent example requests/responses.
5 — Frontend	Pará Amazon map-first experience, selected-area UI, year × month calendar, trends, Sensor Lab and optional CHRONO.	Working interactive website with actual Pará data and transparent measurement labels.
6 — Video / Collaboration Lead	Repository coordination, handoff monitoring, integration QA, scientific-claim review, four-minute video and final submission.	README, scientific limits, integrated demo evidence, final video and submission checklist.

Change-control rule

Member 1 approves geographic boundary, grid and processed data schemas; Member 2 approves model input/output; Member 4 approves public API shape; Member 5 consumes the schema; Member 6 coordinates. All owners affected by a contract change must agree before merging. Use fixtures so frontend/backend can work before the model is finalized.

8. Deadline, release gates and fallbacks

Date	Acceptance milestone
10 Nov	Freeze Pará boundary/version, set shared region_id="BR-PA", benchmark Brazil archive filtering, produce first Pará overlap table and ratio baseline. Backend fixtures and frontend Pará map/calendar working.
11 Nov	Fit/evaluate NB where supported; generate historical Pará predictions; integrate actual queries and API. At least one observed period and one historical supported period must work end to end. Freeze new features at end of day.
12 Nov	Check coordinate clipping, total counts, statuses, model guardrails, map/calendar sync and deploy. Prepare four-minute narrated demo of real science and the study area.
13 Nov	Smoke test, verify model card/source references and submission links; submit before official local cutoff.

Launch gates — all must pass

• User can select at least one supported area inside Pará and move through map → historical calendar → trend → Sensor Lab using real processed data.

• Raw Brazil data never leak into Pará totals unless they pass the frozen state-boundary point-in-polygon test.

• The model report compares a ratio baseline with the selected candidate on held-out data, including geographic performance and limitations.

• A VIIRS-equivalent historical estimate is marked Estimated; post-2012 actual S-NPP is marked Observed; NRT is Provisional; unsupported is null/status, not zero.

• No model-created fake hotspot points; a model operating at 50 km is not presented as a validated 10 km predictive layer.

• API, model and frontend agree on region_id=BR-PA, block_id, year, month, daynight, metric and status definitions.

• Video demonstrates the scientific contribution and functioning app in under four minutes, with honest known limitations.

Fallback ladder — do not break the scientific contract

Blocker	Required response
Full multi-year Brazil archive takes too long	Clip and process representative Pará overlap and historical years first, then expand. Mark other years Not processed.
Negative Binomial is unstable	Use the evaluated ratio baseline, clearly reporting its inability to recover VIIRS-positive/MODIS-zero cases.
Coverage evidence missing	Report preliminary, coverage-unadjusted detected-pixel counts; never claim verified fire-free zeros.
50 km to 10 km prediction request	Keep estimated 50 km display; retain 10 km for observed summaries only.
API deployment instability	Serve precomputed Parquet/static JSON through the agreed same schema as a contingency.
Time pressure	Cut advanced animations, extra regions, new satellite models and extra exports before cutting scientific accuracy or real map/calendar connectivity.

9. Final scope boundaries and team agreement

In scope for the release

- Pará state in Brazil, within an Amazon-centered storytelling context.
- NASA FIRMS Aqua MODIS and S-NPP VIIRS records geographically filtered to Pará.
- Historical VIIRS-equivalent estimated counts where model and observations support them.
- Actual S-NPP observed counts in its era and clearly labeled NRT when optionally present.
- Area selection, historical burning calendar, seasonal charts, model explanation and real validation.

Out of scope until independently supported

- All of Brazil, the full Amazon biome, USA or Argentina as enabled analysis regions.
- Guaranteeing a complete, validated 2002–2026 monthly calendar before ingestion and holdout checks.
- Terra-only 2000–2001 historical conversion without separate calibration.
- Live fire alerts, fire spread prediction, unique fire incident counts or burned-area estimates.
- Unvalidated cloud/coverage correction, kiln masking, NOAA-20 substitution, confidence intervals or spatial downscaling.
- Claims that a 50 km or Pará-calibrated conversion automatically works at every 10 km location or outside Pará.

Single source of truth

The six-person team builds one Pará-based FireChronicle release. One frozen study boundary, one grid function, one model output contract, one backend response schema, and one visible scientific story. Genuine NASA data and transparent validation come before optional features. No invented values will be used to make a calendar look complete.

Acknowledgment: The collaboration lead records the six members’ acceptance in the shared GitHub issue or team channel. Only approved changes to scientific assumptions, geography, or interface fields supersede this rulebook.